Development of Lignocellulose-Based Hydrogel Adhesives for Medical Applications

Research Experience

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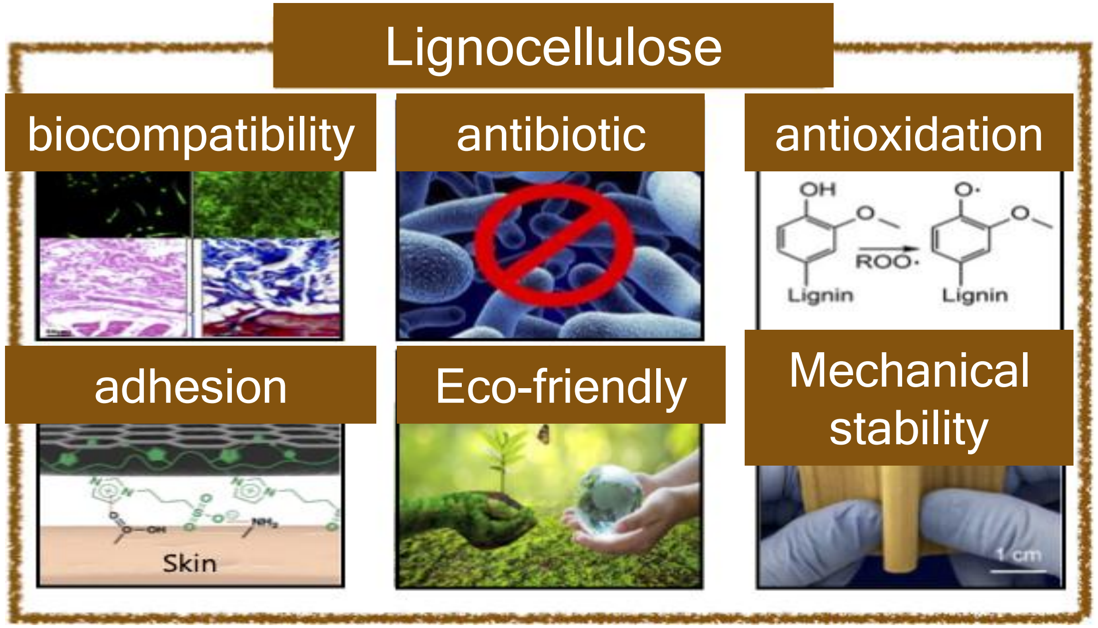
Project - Development of Lignocellulose-Based Hydrogel Adhesives for Medical Applications

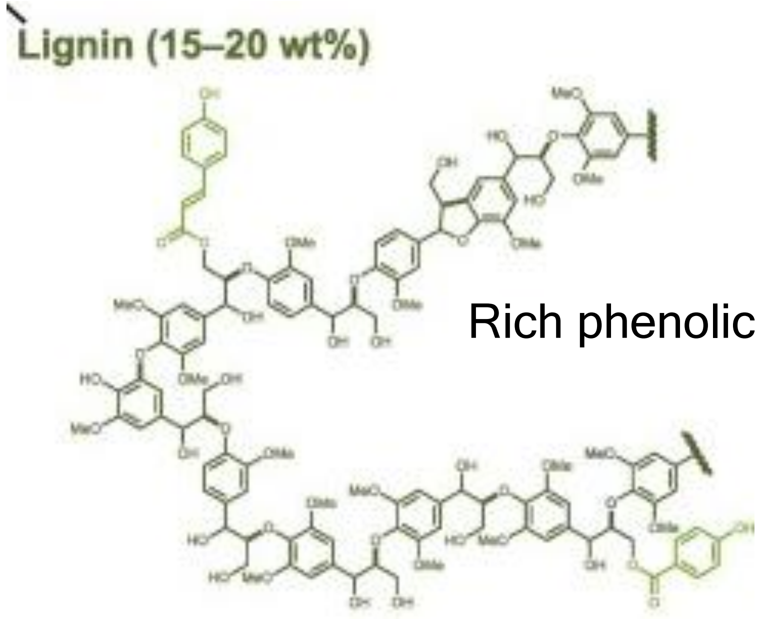
Requirements for Biomedical Adhesive Materials

- High adhesion strength with minimal mechanical burden on attached tissues
- Dimensional stability in wet environments (non-swelling, non-dissolving behavior)
- Biocompatibility



Valorization of Lignocellulosic Biomass

- Abundant, underutilized lignocellulosic biomass
  → high-value functional biomaterials



Rich phenolic groups!

➔ Engineering physically interacting polymer networks for high-adhesion and tissue-adaptive biomedical hydrogels

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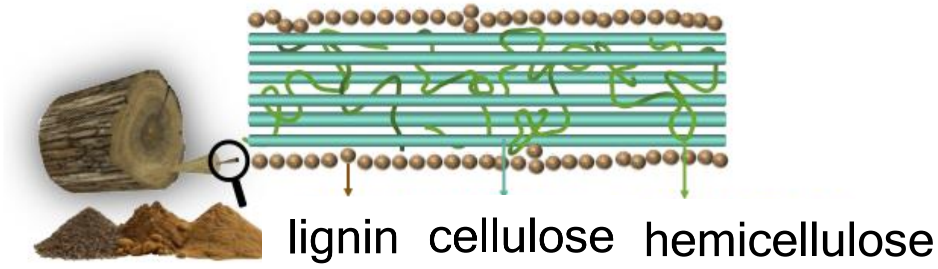
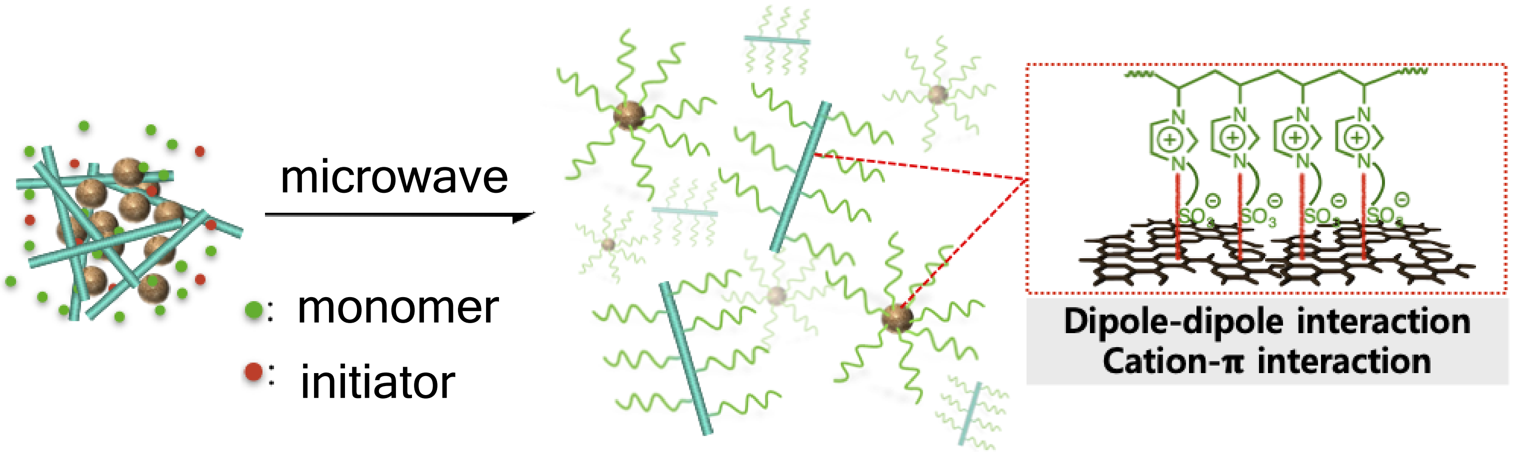
Lignin + Sulfobetaine vinylimidazolium (SBVI) (zwitterionic monomer) $\xrightarrow[\text{Polymerization of SBVI and Lignin}]{}$ Lignogel

Lignocellulose



- hydrogen bonding
- π-π stacking
- cation-π interaction
- ionic interaction



Other Zwitterionic Monomers

Lignogel with SBMA

Gelation X

Lignogel with MPC

Swelling / Poor Adhesion

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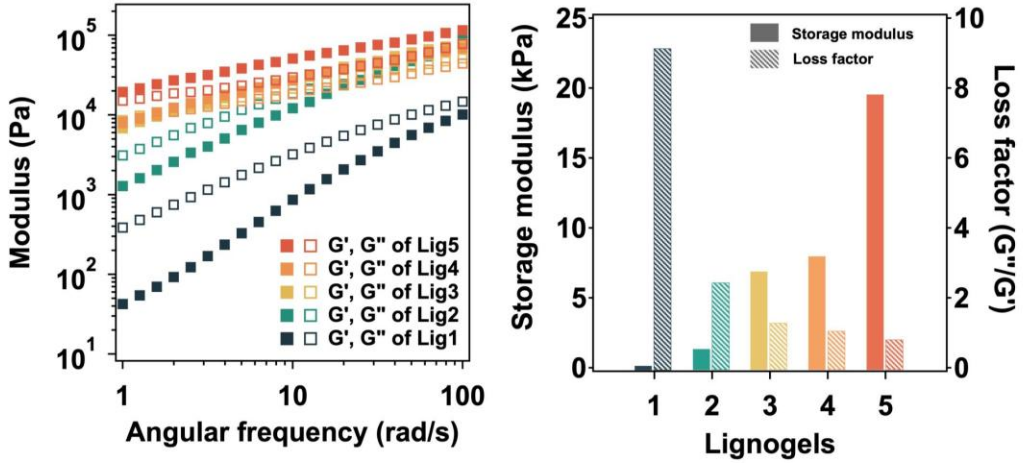
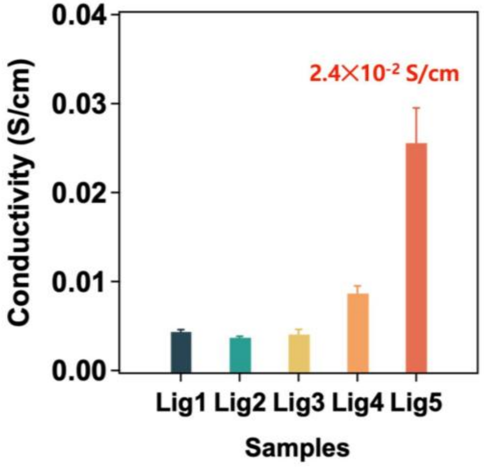
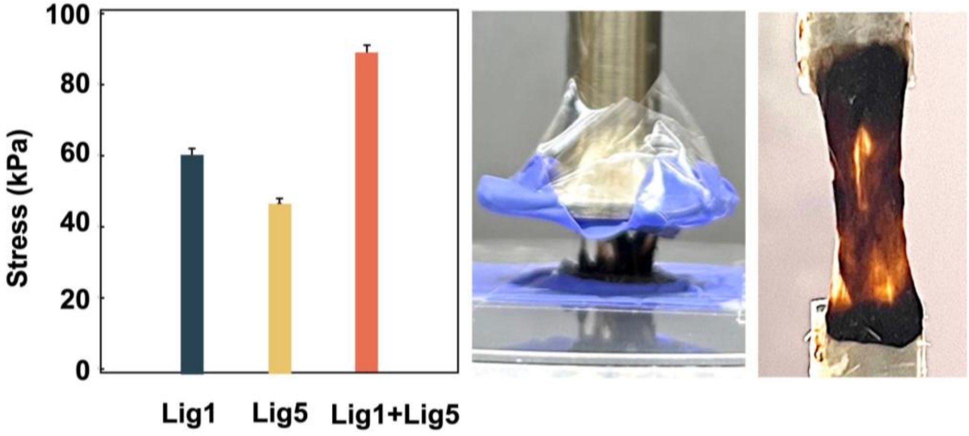
Viscoelastic Tunability



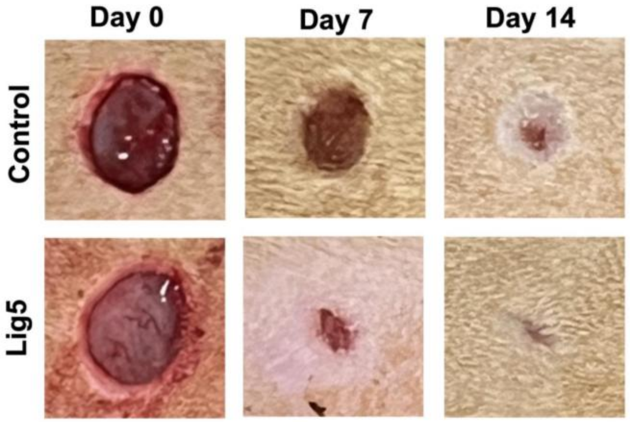
Conductivity



Adhesion



Wound-Healing Capability



➔ Wet-stable, biocompatible, high-adhesion lignocellulosic hydrogel adhesive platform

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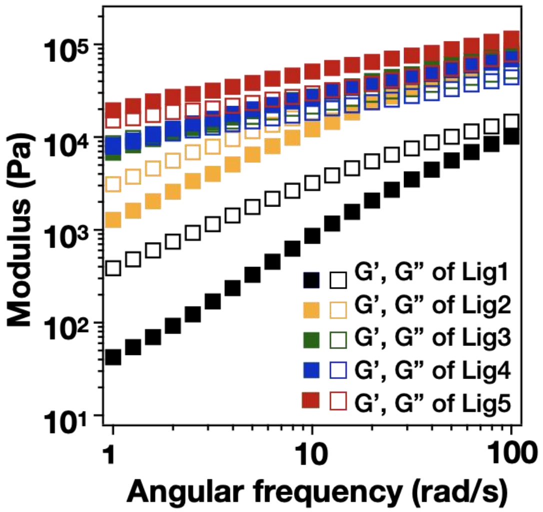
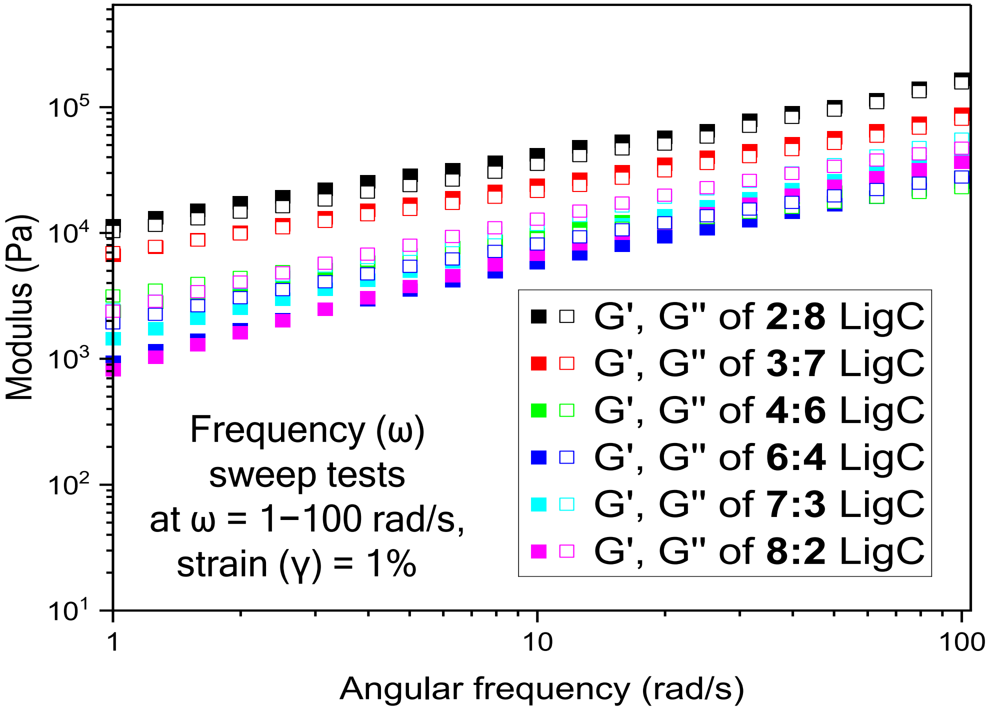


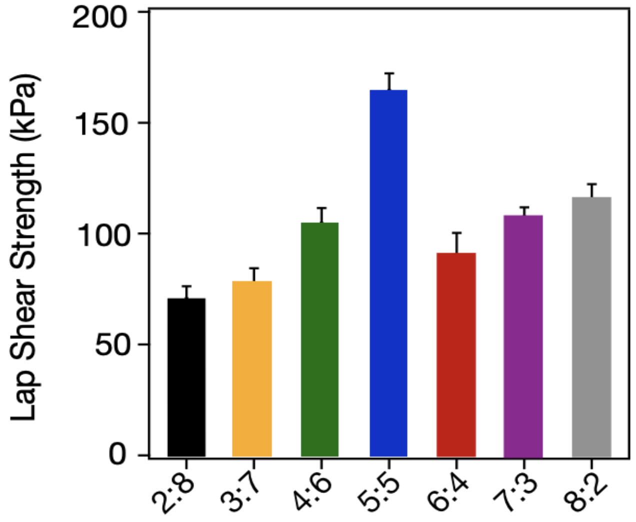
Lig1	Lig5
Free chain ↑	Free chain ↓
Adhesion ↑	Adhesion ↓
Cohesion ↓	Cohesion ↑

Blending Lig1 and Lig5

LigC

Adhesion ↑, Cohesion ↑





50 mm/min
Porcine skin
: 10 x 10 mm